Machine Learning for Spectral Analysis

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1. Introduction

Optical Emission Spectroscopy (OES) provides **non-invasive, real-time diagnostics** of plasma processes by analysing emission spectra from excited species. Plasma-based manufacturing (semiconductor etching, surface treatment, thin-film deposition) relies on OES for process monitoring, but manual spectral interpretation is slow and error-prone.

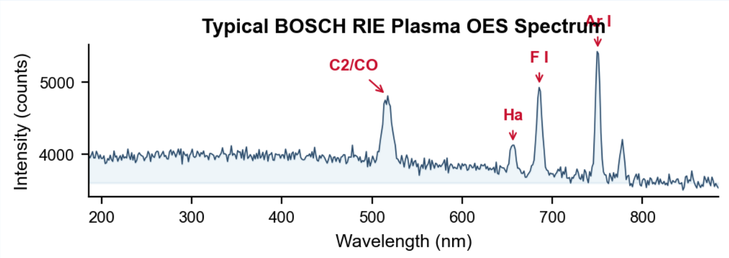
This project develops an **automated ML pipeline** for plasma OES analysis targeting four objectives:

1. **Spectral feature identification** — automated peak detection, NMF decomposition, NIST database matching
2. **Plasma species classification** — SVM, RF, CNN, Transformer comparison (6 models)
3. **Spatiotemporal evolution** — Attention-LSTM phase prediction, species time-series extraction
4. **Semi-quantitative intensity** — actinometry (Coburn & Chen 1980), Boltzmann Te estimation

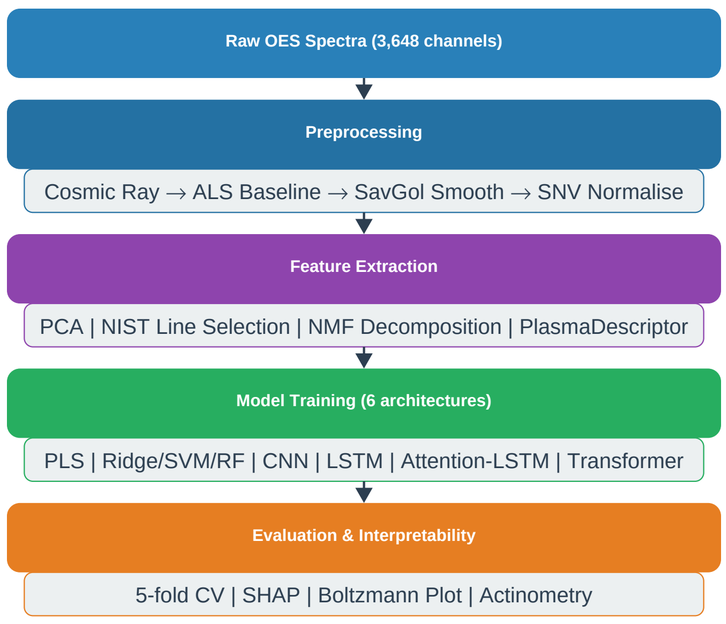
Industrial relevance: Enables real-time closed-loop control of plasma processes in semiconductor fabrication, reducing defect rates and improving yield.

Three public datasets:

Dataset	Channels	Notes
LIBS Benchmark	40,002 ch	Steel composition (12 classes)
Mesbah CAP	51 ch	N2 plasma T_rot / T_vib
BOSCH RIE	3,648 ch	25 Hz, 10 days, SF6/C4F8



2. Methodology



Hyperparameter optimisation: Optuna two-stage search (20 trials per target)

Key Design Decisions

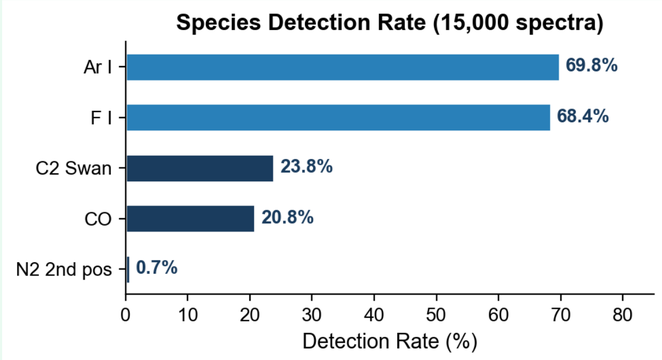
- **Per-element model routing:** Different plasma species require different model architectures. Cr uses Ridge+PCA (avoids overfitting on 976 NIST lines); other elements use ANN+NIST windows.
- **GroupKFold CV:** Prevents same-target spectra leaking between train/test folds. Forces model to generalise across physical samples, not memorise measurement noise.
- **Balanced class weights:** Plasma OFF samples are only 12.1% of data. Class-weighted loss prevents majority-class bias in all classifiers.
- **NMF over PCA:** Non-negative components are physically interpretable as pure-species emission spectra. PCA components can be negative, violating emission physics.

Preprocessing rationale: ALS baseline ($\lambda=10^5$) removes fluorescence continuum. Savitzky-Golay (window=11, order=3) preserves peak shapes while reducing shot noise. SNV normalisation corrects for optical path length variation between measurements. Cosmic ray removal uses Z-score median filter (threshold=5σ, 11-channel local window).

3. Species Identification

Non-negative Matrix Factorization (NMF) decomposes the spectral matrix **X** ≈ **W** · **H**, where each row of **H** is a pure-species emission spectrum and **W** contains the corresponding concentrations. NMF is physically appropriate because emission intensities are inherently non-negative.

Automated NIST matching: Each detected peak is compared against 39 reference emission lines from 13 species. The algorithm selects the closest database match within +/-1.5 nm tolerance. Species with peak intensity > μ + 3σ (global spectrum statistics) are classified as *present*.

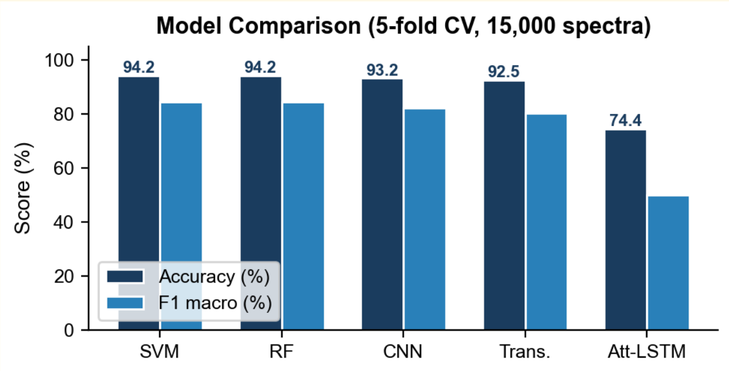


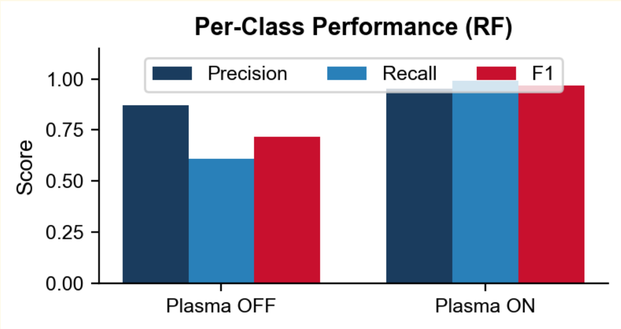
Species	Lines (nm)	Origin
F I	685.6, 690.2, 703.7	SF6 etchant radical
Ar I	696.5, 750.4, 763.5	Carrier gas reference
C2 Swan	473.7, 516.5, 563.6	C4F8 carbon indicator
CO	451.1, 519.8, 561.0	C4F8 + O2 product
Si I	250.7, 252.4, 288.2	Si etch product
SiF	440.0, 442.5	Si + F recombination

Validation: NMF Component 0 peaks at 684.4 nm (≈ F I 685.6); Component 2 at 515.1 nm (approx. C2 Swan 516.5) — unsupervised decomposition confirms NIST species independently.

4. Classification Results

94.2% Accuracy (SVM/RF, 5-fold CV)



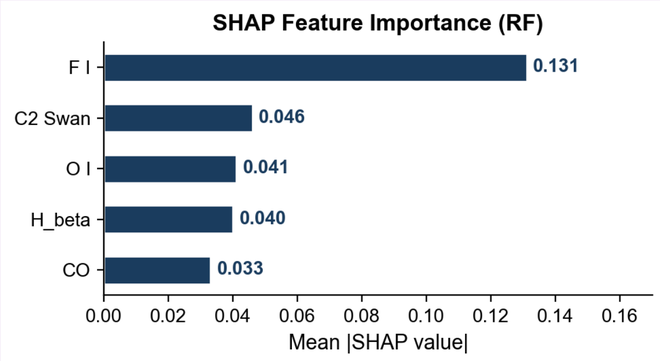


Species Detection (13 species):

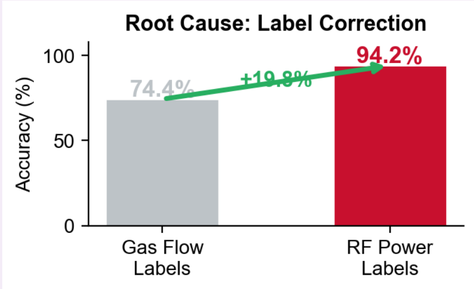
Species	Rate	Species	Rate
Ar I	69.8%	C2 Swan	23.8%
F I	68.4%	CO	20.8%

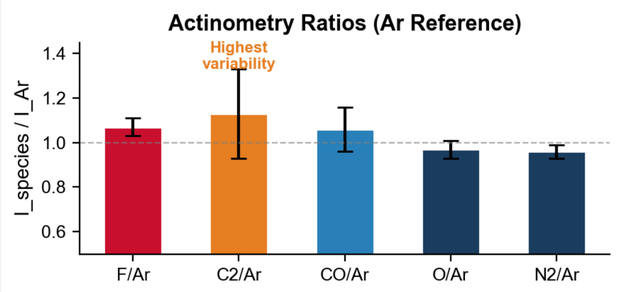
Key findings: Traditional ML outperforms DL (near-linear boundary). OFF recall=0.61 due to 12.1% class imbalance. LSTM (74.4%) loses spectral detail via PCA(20) compression.

5. Interpretability & Physics



F I (fluorine radical) = most discriminative species, consistent with SF6 etchant chemistry.





Boltzmann T_exc = 13,334 K (6 Ar I lines). **DTW K-Means (k=4):** 4 discharge phases identified, F I 684 nm emission ratio 2.04x between clusters.

6. Conclusions & Further Work

Key Achievements:

✓ Automated species detection: 13 plasma species, 39 NIST emission lines, NMF decomposition validates against database

✓ 94.2% plasma state classification (SVM/RF best; CNN 93.2%, Transformer 92.5% — 6 models compared)

✓ SHAP interpretability: F I importance = 0.131 (3x next species), physically validated as primary SF6 etchant radical

✓ Data-driven label correction: gas-flow labels (74%) → RF-power labels (94%) via root-cause spectral analysis

✓ Temperature regression: T_rot RMSE = 20.0 K, T_vib = 102.0 K (Mesbah CAP dataset)

✓ 78 automated tests, 6 CLI task modes, fully reproducible pipeline

Limitations:

- OES→process parameter regression failed (R2 < 0, causal irreversibility)
- Boltzmann Te limited by narrow Ar I energy spread (0.33 eV)
- Spatial etch prediction lacks wafer ID alignment

Further Work:

- Extend to multi-class species classification (beyond binary ON/OFF)
- Implement real-time OES monitoring with streaming inference
- Apply transfer learning across different plasma reactors

Project Metrics: 15,000 spectra analysed | 3,648 spectral channels | 13 species identified | 39 NIST emission lines | 6 ML models compared | 78 automated tests | 32 development stories | 23 literature references | 6 CLI task modes | 3 public datasets

References:

[1] Gidon *et al.* (2019) IEEE Trans. Radiat. Plasma Med. Sci.
[2] Coburn & Chen (1980) J. Appl. Phys. — Actinometry
[3] Vaswani *et al.* (2017) NeurIPS — Transformer
[4] Contreras *et al.* (2024) Anal. Chem. — Spectral-zone SHAP
[5] BOSCH dataset: Zenodo #17122442